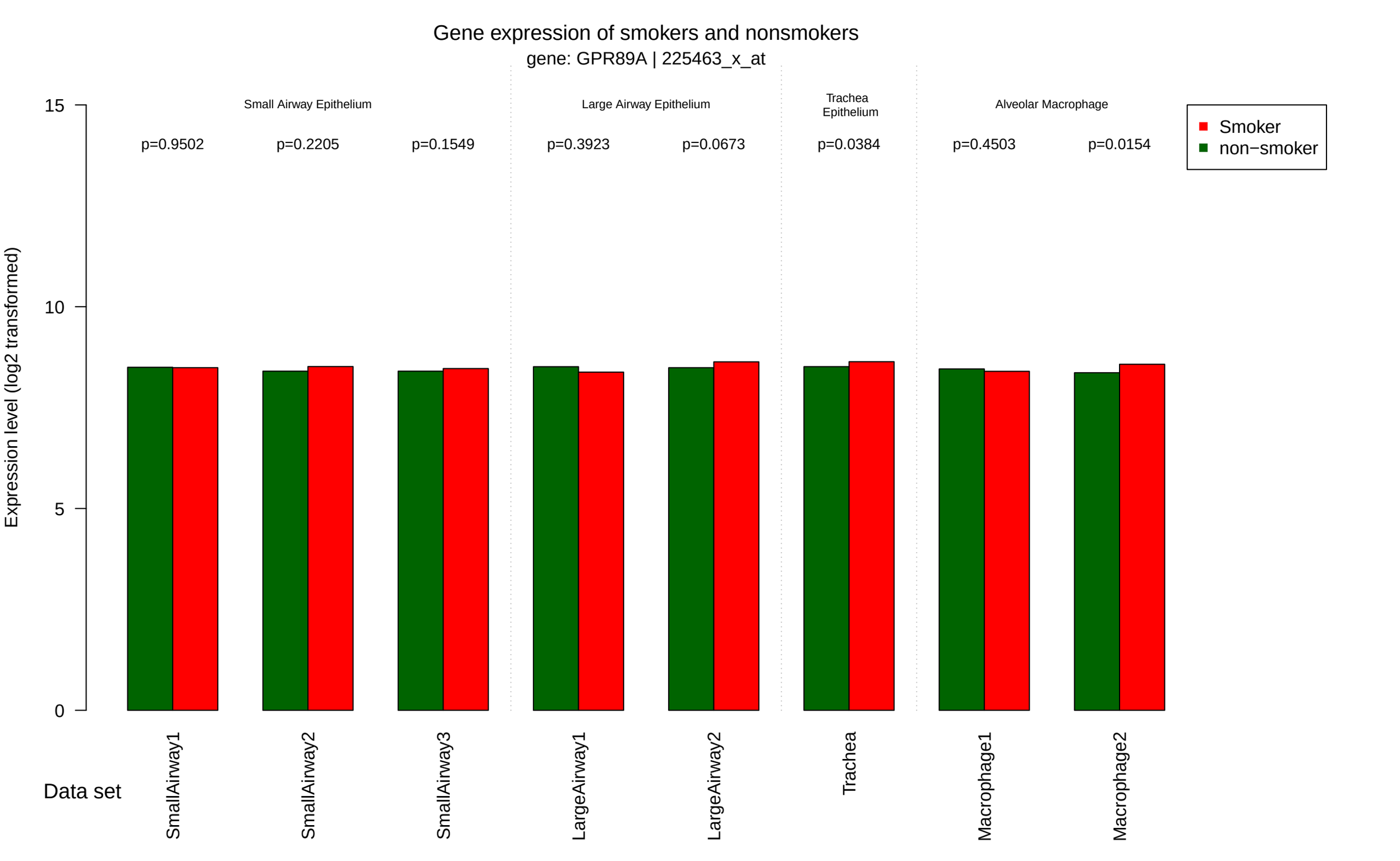
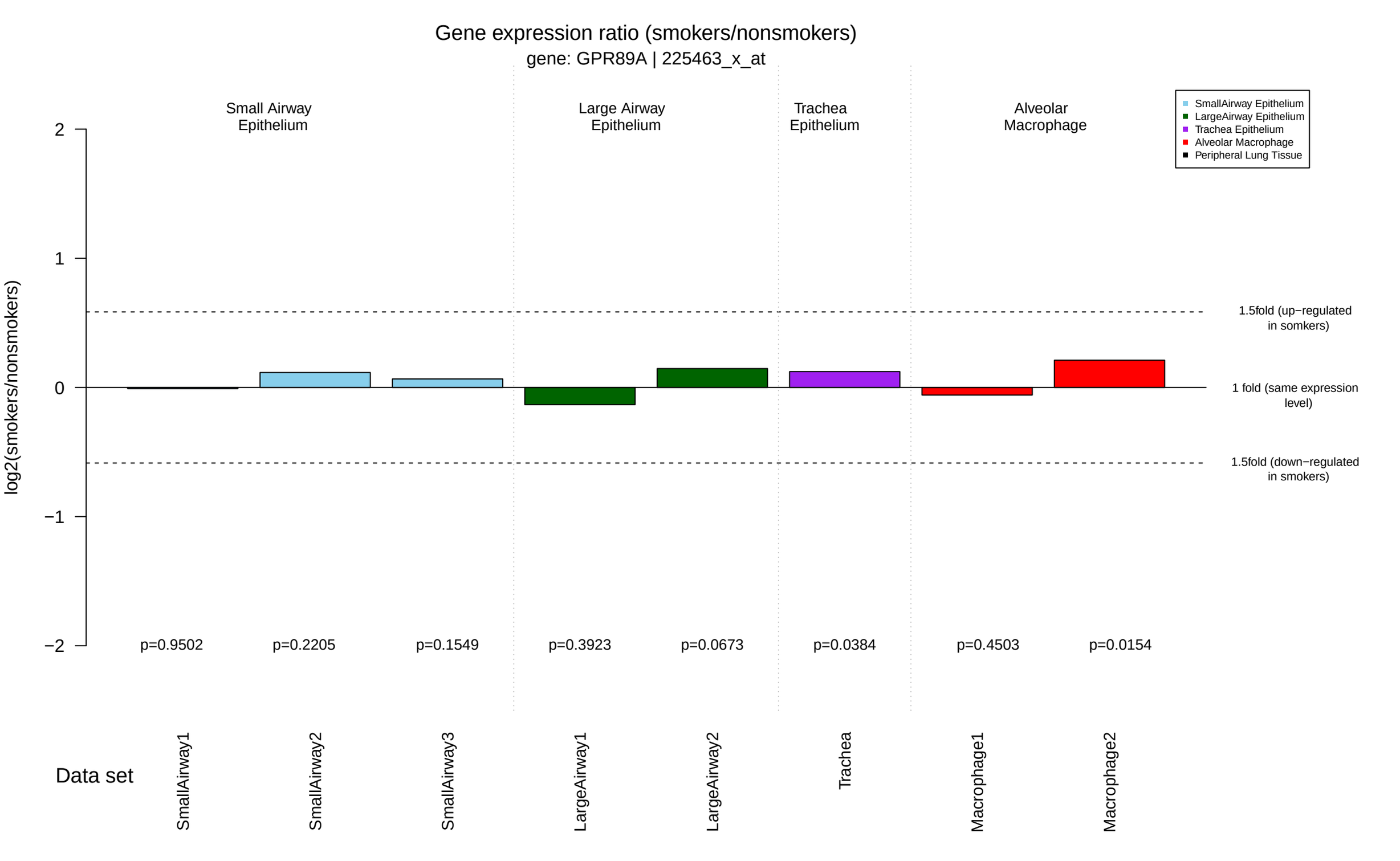
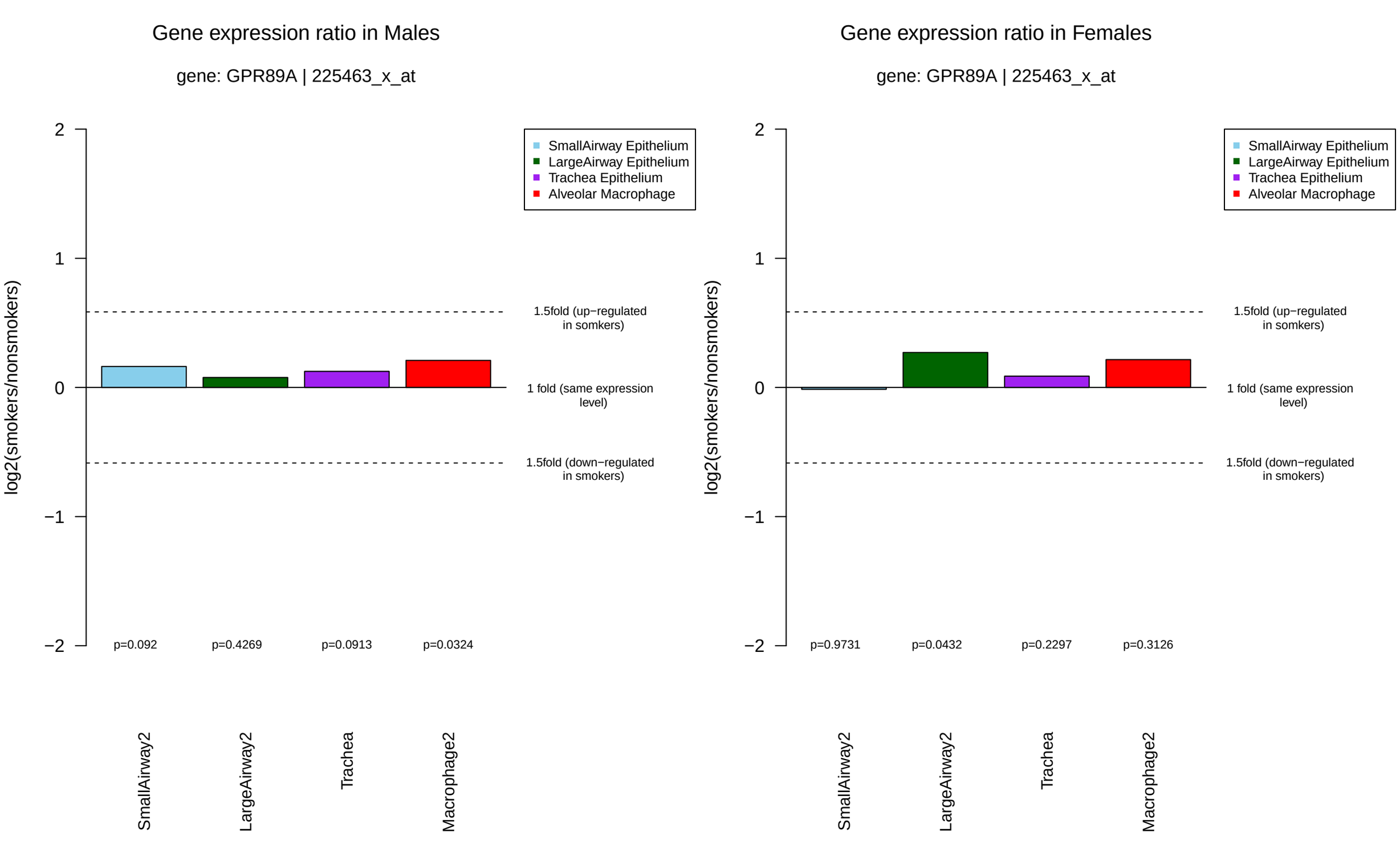
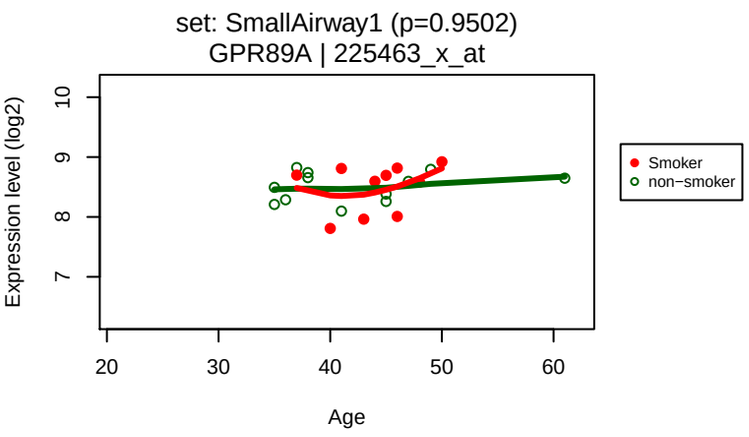
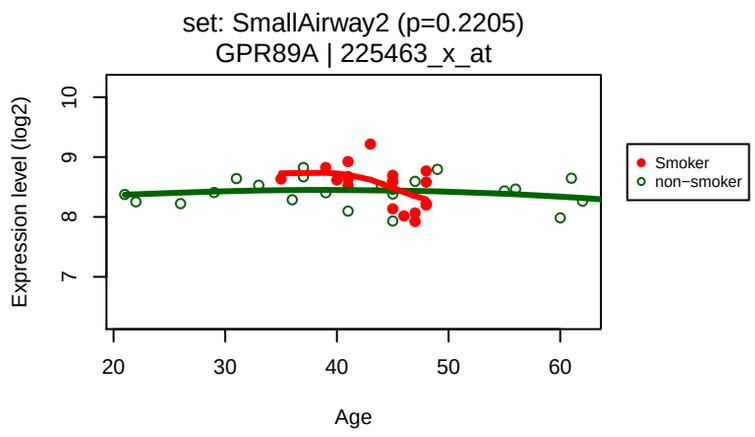
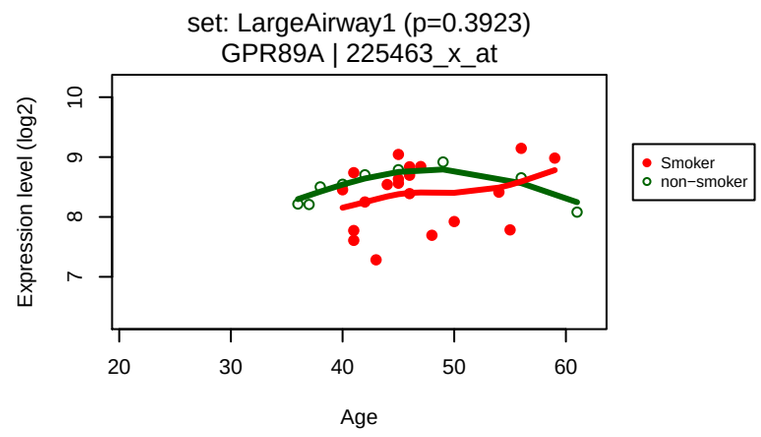

Gene expression of smokers and nonsmokers
gene: GPR89A | 225463_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9502
p=0.2205
p=0.1549
p=0.3923
p=0.0673
p=0.0384
p=0.4503
p=0.0154
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

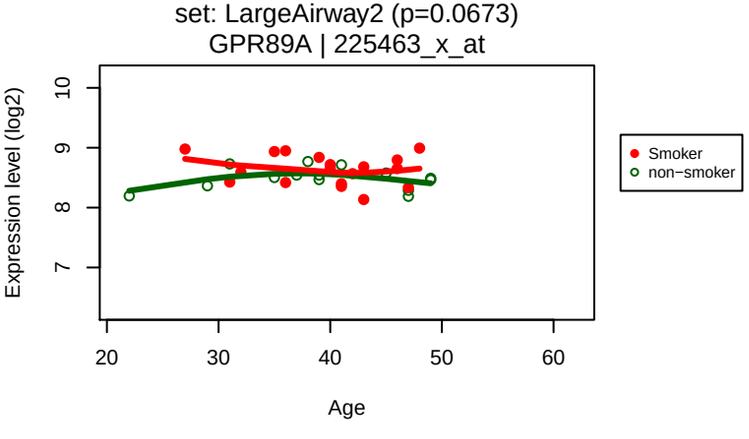

Gene expression ratio (smokers/nonsmokers)
gene: GPR89A | 225463_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9502
p=0.2205
p=0.1549
p=0.3923
p=0.0673
p=0.0384
p=0.4503
p=0.0154
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

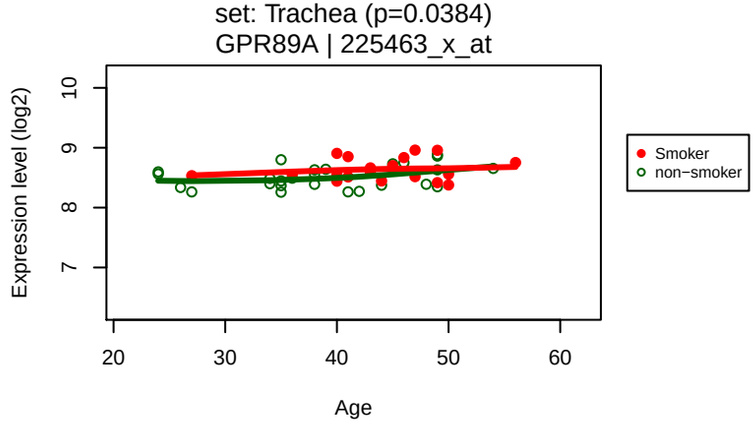

Gene expression ratio in Males
gene: GPR89A | 225463_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.092
p=0.4269
p=0.0913
p=0.0324
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPR89A | 225463_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9731
p=0.0432
p=0.2297
p=0.3126
SmallAirway2
LargeAirway2
Trachea
Macrophage2

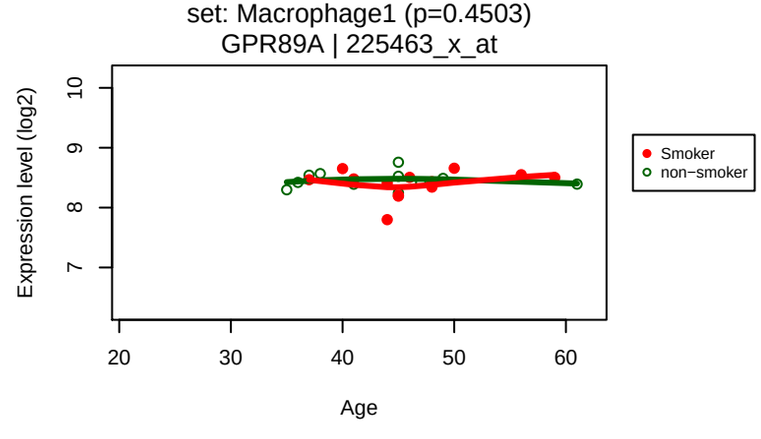

set: SmallAirway1 (p=0.9502)
GPR89A | 225463_x_at
Expression level (log2)
Age
Smoker
non−smoker

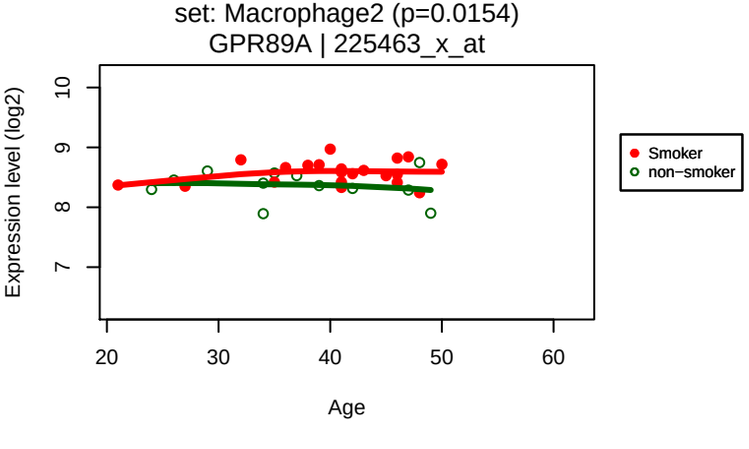

set: SmallAirway2 (p=0.2205)
GPR89A | 225463_x_at
Expression level (log2)
Age
Smoker
non−smoker

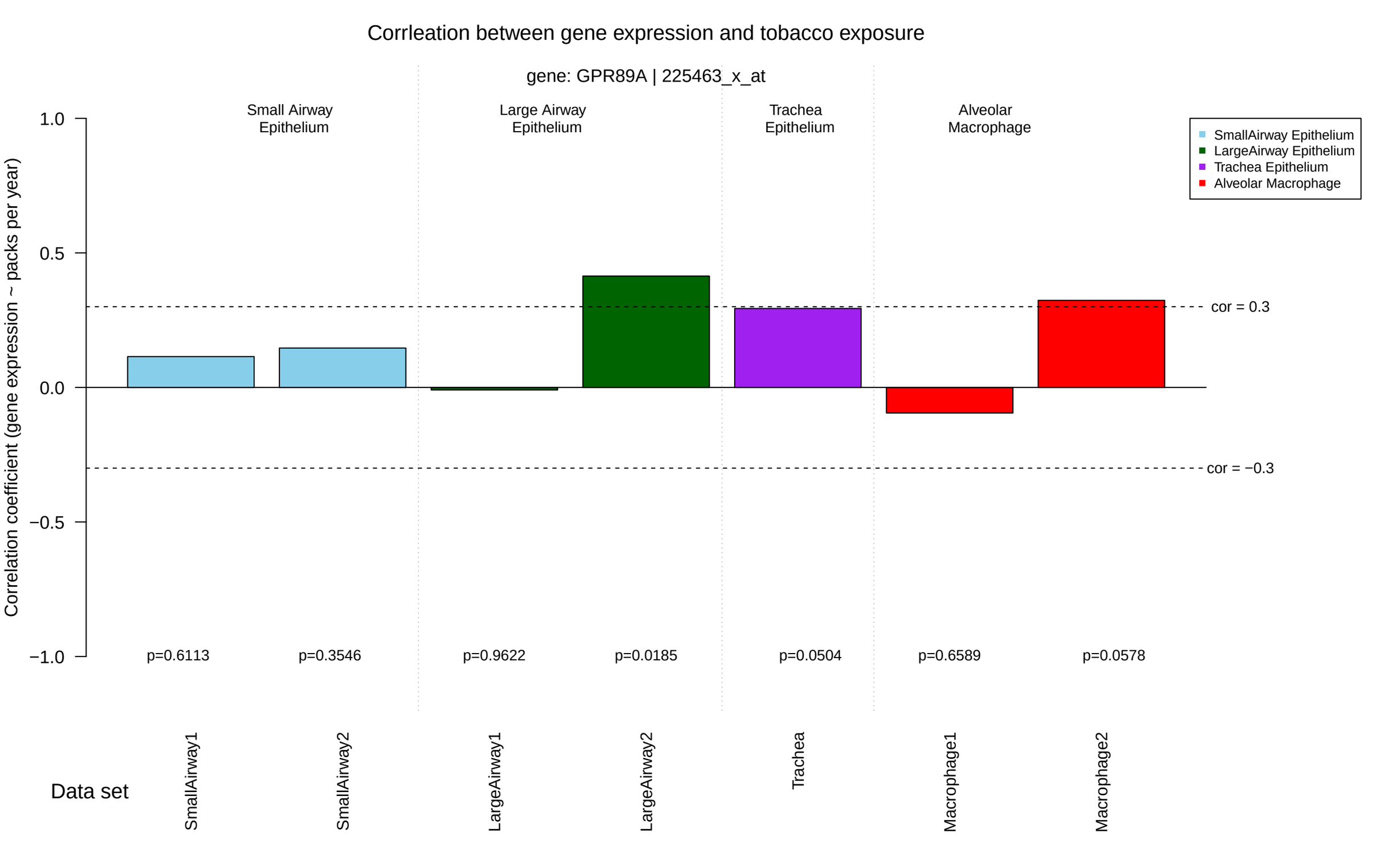

set: LargeAirway1 (p=0.3923)
GPR89A | 225463_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0673)
GPR89A | 225463_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0384)
GPR89A | 225463_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4503)
GPR89A | 225463_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0154)
GPR89A | 225463_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPR89A | 225463_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6113
p=0.3546
p=0.9622
p=0.0185
p=0.0504
p=0.6589
p=0.0578
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2